Digital Inclusion and Circular Economy Behavior in Indian States:
A Digital Circularity Index and Explainable XGBoost Analysis

21CSP302L - PROJECT

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ABSTRACT

India is simultaneously expanding digital infrastructure and pursuing circular economy (CE) goals, yet the relationship between these two agendas remains underexplored at the state level. This study integrates seven official government data sources to construct a Digital Circularity Index (DCI) for 16 Indian states. The analysis combines Ordinary Least Squares (OLS) regression, robustness testing, mediation analysis, threshold analysis, leak-free nested cross-validation, SHAP-based explainability, and K-Means clustering.

Across every method tried, states with stronger digital access consistently show better circular economy outcomes. The OLS model yields $r = 0.781$ ($p < 0.001$, $R^2 = 0.610$), and the leak-free nested-CV XGBoost still reaches $R^2 = 0.511$. The association survives non-parametric and permutation-based checks, while tele-density emerges as a stronger driver than digital payments. K-Means clustering separates the 16 states into two distinct typologies — Digital Leaders and Emerging States — with a 2.5× DCI gap.

The findings suggest that digital inclusion and sustainability policy should be treated as interconnected agendas rather than separate tracks. The work contributes a reproducible state-level composite index, cross-method statistical and machine learning validation, and state-specific policy interpretation through SHAP and clustering.

Keywords: Circular economy, Digital Circularity Index (DCI), Digital inclusion, XGBoost, SHAP, Policy analytics, India.

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ABBREVIATIONS

Abbreviation	Full Form
CE	Circular Economy
CEBS	Circular Economy Behavior Score
CPCB	Central Pollution Control Board
DAS	Digital Access Score
DCI	Digital Circularity Index
EDA	Exploratory Data Analysis
EPR	Extended Producer Responsibility
HC3	Heteroskedasticity-Consistent Standard Errors (type 3)
ICT	Information and Communication Technology
IEEE	Institute of Electrical and Electronics Engineers
LOO-CV	Leave-One-Out Cross-Validation
ML	Machine Learning
NITI Aayog	National Institution for Transforming India
NPCI	National Payments Corporation of India
NSDP	Net State Domestic Product
OLS	Ordinary Least Squares
PCA	Principal Component Analysis
RBI	Reserve Bank of India
RF	Random Forest
RMSE	Root Mean Squared Error
SDG	Sustainable Development Goal
SHAP	Shapley Additive Explanations

SVR	Support Vector Regression
TRAI	Telecom Regulatory Authority of India
UPI	Unified Payments Interface
VIF	Variance Inflation Factor
XGBoost	Extreme Gradient Boosting

CHAPTER 1 — INTRODUCTION

1.1 Introduction to Project

Since the launch of the Digital India programme in 2015, India has dramatically expanded broadband penetration, mobile connectivity, and the Unified Payments Interface (UPI) across a population of 1.4 billion people. Monthly UPI transactions now exceed 14 billion, and national tele-density is above 85%. In parallel, India generates roughly 62 million tonnes of waste annually and ranks among the world's top three e-waste producers; NITI Aayog's circular economy (CE) strategy now spans 11 priority sectors.

This project, titled "Digital Inclusion and Circular Economy Behavior in Indian States", is a data-driven research initiative that investigates whether digital inclusion drives circular economy outcomes at the state level. The work integrates seven official government data sources — TRAI, RBI, NITI Aayog, NPCI, CPCB, Census of India, and Dataful — covering 16 Indian states with 100% verified observations, and constructs three novel composite indices: a Digital Access Score (DAS), a CE Behavior Score (CEBS), and the Digital Circularity Index (DCI).

The analytical pipeline combines classical statistics (OLS, non-parametric tests, bootstrap, mediation, piecewise regression) with machine learning (seven LOO-CV models, XGBoost with nested CV) and model-agnostic explainability (SHAP), and closes with K-Means clustering of state typologies. The final deliverables include 22 publication-ready figures, a defense-ready IEEE conference paper, and a reproducible code base.

1.2 Problem Statement and Description

India is pursuing two major national agendas in parallel — rapid digitization under Digital India and a transition to a circular economy under Swachh Bharat and NITI Aayog's 11-sector CE strategy. Research on Digital India has focused mainly on growth, employment, and financial inclusion, while CE research has stayed centered on waste-management technology and Extended Producer Responsibility (EPR) policy. Very little state-level work has asked whether stronger digital access is actually associated with better environmental performance.

The core research problem addressed by this project is therefore:

> ***"Does digital inclusion — measured through tele-density, UPI adoption, and economic capacity — predict circular-economy behavior across Indian states, and if so, through what mechanisms?"***

To operationalise this problem, three testable hypotheses are defined:

- H1 — States with higher Digital Access Scores will also have higher CE Behavior Scores.
- H2 — UPI adoption mediates this relationship, i.e., digital payments are the channel through which digital access affects CE behavior.
- H3 — There is a tele-density threshold beyond which CE behavior improves sharply.

1.3 Motivation

Three motivations drive this work:

1. Policy coherence. Digital India and Swachh Bharat are usually treated as independent policy tracks. If digital infrastructure measurably lifts CE outcomes, then connectivity investment is not only a telecom intervention but also a sustainability intervention — a finding with direct implications for budget allocation.
2. State-level evidence gap. State SDG scores range widely from 57 to 79, Bihar's tele-density is around 55.8% while Kerala exceeds 120%, yet no composite index ties digital access to circular economy outcomes at the sub-national level. This project fills that gap.
3. Methodological rigor under small-n. With only 16 states in the clean merged dataset, naive modeling overfits. The project demonstrates a defensible workflow that combines bootstrap, LOO-CV, nested CV, and SHAP, which is itself a reusable template for other small-n, policy-relevant research questions.

1.4 Sustainable Development Goal of the Project

The project directly contributes to multiple United Nations Sustainable Development Goals (SDGs) adopted as part of the 2030 Agenda:

SDG	Relevance to Project

SDG 9	Industry, Innovation and Infrastructure — tele-density and UPI act as proxies for digital infrastructure and innovation capacity.
SDG 11	Sustainable Cities and Communities — urbanisation and waste handling directly inform the CEBS component.
SDG 12	Responsible Consumption and Production — e-waste processed per capita is a core input to the CE Behavior Score.
SDG 13	Climate Action — improved CE behavior reduces emissions from landfill and virgin-material extraction.
SDG 17	Partnerships for the Goals — the study uses data from seven public institutions and supports cross-ministry coordination.

The central policy message — that digital inclusion and circular economy agendas should be treated as connected rather than separate — is consistent with NITI Aayog's SDG India Index 2023-24 direction and the UN's integrated SDG framework.

CHAPTER 2 — LITERATURE SURVEY

2.1 Overview of the Research Area

The study sits at the intersection of three research streams: (1) digital inclusion and ICT-for-development, (2) circular economy and industrial ecology, and (3) data-driven policy analytics. Between 2015 and 2024, India pushed tele-density beyond 85% nationally, with urban pockets crossing 130%. Since 2016, UPI has grown into the world's largest real-time digital payments system. Yet regional disparities remain stark — Bihar sits around 55.8% tele-density while Kerala exceeds 120%.

On the circular economy side, the Ellen MacArthur Foundation popularised the framework, and Geissdoerfer et al. (2017) later distinguished it more clearly from the broader idea of sustainability. In India, NITI Aayog translated that agenda into an 11-sector strategy, but implementation remains uneven: the country generates roughly 1.71 million tonnes of e-waste annually, only part of which enters formal recycling channels, and state SDG scores still range from 57 to 79.

Avgerou and Heeks argue that ICTs reshape development through multiple channels — governance, markets, information flow, and institutional functioning. If that view holds, digital access should plausibly affect CE behavior through several mechanisms rather than one isolated pathway. However, state-level empirical evidence connecting Digital India with circular-economy outcomes remains limited — the gap this study addresses.

2.2 Existing Models and Frameworks

Table 2.1 summarises the most relevant models and frameworks reviewed from the literature, grouped by their analytical purpose.

Model / Framework	Area	Key Contribution	Limitation
Ellen MacArthur Foundation (2013)	Circular Economy	Conceptual 'butterfly' framework separating biological and technical	Qualitative, not linked to digital infrastructure.

		cycles.	
Geissdoerfer et al. (2017)	Circular Economy vs. Sustainability	Systematic review that separates CE from the broader sustainability idea.	Framework-level, no empirical state-level model.
NITI Aayog (2021, 2024)	CE Strategy / SDG India Index	11-sector CE roadmap and sub-national SDG Index (Kerala 79 vs. Bihar 57).	Policy document; no linkage model to digital inclusion.
Avgerou (2010); Heeks (2017)	ICT4D	ICTs act on governance, markets, and institutions — multi-channel theory.	Theoretical, not specialised to Indian CE outcomes.
Baron & Kenny (1986); Sobel test	Mediation Analysis	Classical four-step approach for testing mediator variables.	Low power with small samples; sensitive to distributional assumptions.
Lundberg & Lee (2017)	SHAP	Unified, game-theoretic framework for model-agnostic explainability.	Requires careful handling for small n; expensive for high-dimensional data.
XGBoost (Chen & Guestrin)	Gradient Boosting	Regularised tree ensembles with strong performance on tabular data.	Needs nested CV when tuning on small n.

2.3 Limitations Identified from Literature Survey (Research Gaps)

The literature review surfaces four concrete gaps that this project is designed to close (Table 2.2).

ID	Gap	How This Project Closes It
G1	No state-level composite linking digital inclusion and circular economy exists for India.	Build a Digital Circularity Index (DCI) from 7 verified sources.
G2	Most Digital India studies stop at access/adoption; CE studies stop at waste-flow	Model DCI = f(DAS, CEBS) jointly, not in

	metrics.	isolation.
G3	Small-n, policy-relevant studies in India rarely pair OLS with leak-free ML validation.	Combine OLS + 7-model LOO-CV + nested-CV XGBoost + SHAP.
G4	Limited state typology work that blends econometrics with clustering.	Use K-Means (k=2, silhouette-selected) + PCA to profile Digital Leaders vs. Emerging States.

2.4 Research Objectives

Following the research gaps, the objectives of the project are expressed as five Epics that also structure the product backlog in Section 2.5:

4. Objective O1 — Build a unified, verified, state-level dataset from seven official government sources, with explicit provenance and zero imputation.
5. Objective O2 — Construct three composite indices (DAS, CEBS, DCI) using transparent min–max normalisation and justifiable weight schemes.
6. Objective O3 — Test the primary hypothesis H1 using OLS and non-parametric methods, plus bootstrap, HC3 robust errors, permutation tests, and weight sensitivity analysis.
7. Objective O4 — Validate the relationship using out-of-sample predictive machine learning, with LOO-CV across seven models and leak-free nested LOO-CV for XGBoost.
8. Objective O5 — Explain individual state behavior through SHAP and identify natural state typologies through K-Means clustering + PCA, closing the loop with policy implications.

2.5 Product Backlog (Key User Stories with Desired Outcomes)

The product backlog organises the research objectives into epics and actionable user stories. Each story has a clear 'desired outcome' column that doubles as its acceptance criterion.

Story ID	Epic	User Story	Desired Outcome / Acceptance

E1-US1	Epic 1 — Data Engineering	As a researcher, I want to merge 7 government sources via inner-join so I can work with one traceable dataset.	Unified 16-state × 15-variable CSV, 0 missing values.
E1-US2	Epic 1 — Data Engineering	As a researcher, I want normalised composite indices (DAS, CEBS, DCI) with documented weights.	Min–max normalised variables + three composite columns with weight justification.
E2-US1	Epic 2 — EDA	As a researcher, I want 8+ publication-ready EDA figures so I can spot patterns before hypothesis testing.	Figures 1–8 generated: heatmap, scatter, DCI ranking, distributions, boxplots, UPI trends, regression panels, UPI vs SDG.
E3-US1	Epic 3 — Statistics	As a researcher, I want to test H1 with OLS + full diagnostics.	OLS: $\beta_1 = 0.551$, $R^2 = 0.610$, $p < 0.001$; Breusch-Pagan, Durbin-Watson, Cook's D reported.
E3-US2	Epic 3 — Statistics	As a researcher, I want robustness checks so reviewers cannot dismiss the finding on small-n or distributional grounds.	HC3 $p = 1.1\times10^{-5}$, permutation $p = 0.0001$, bootstrap CI [0.648, 0.979].
E3-US3	Epic 3 — Statistics	As a researcher, I want to test H2 (UPI mediation) and H3 (tele-density threshold).	Sobel $z = -0.241$ (no mediation); piecewise breakpoint at 118.77% with $R^2 = 0.723$.
E4-US1	Epic 4 — ML Validation	As a researcher, I want 7 ML models under LOO-CV to confirm the signal survives out-of-sample.	Full-model LOO-CV: Lasso $R^2 = 0.568$; digital-only: ElasticNet $R^2 = 0.364$.
E4-US2	Epic 4 — ML Validation	As a researcher, I want nested LOO-CV to remove hyperparameter leakage.	Nested-CV XGBoost $R^2 = 0.511$, $p = 0.002$.

E5-US1	Epic 5 — Explainability	As a researcher, I want SHAP values so I can explain state-level behavior.	Global importance: tele-density (5.01) and NSDP (4.88) top ranks; per-state waterfall plots for Haryana and Bihar.
E5-US2	Epic 5 — Clustering	As a policy analyst, I want data-driven state typologies.	K-Means k = 2: Digital Leaders (DCI 46.7) vs. Emerging States (DCI 18.3); 2.5× gap.

2.6 Plan of Action (Project Road Map)

The six-week sprint plan, summarised in Table 2.4, maps directly to the five epics above and to the two sprint windows that structure Chapter 3.

Week	Dates	Sprint / Epic	Deliverables
Week 1	01 Feb – 08 Feb 2026	Sprint I / Epic 1	Source data from TRAI, RBI, NITI Aayog, NPCI, CPCB, Census, Dataful. Merge via inner-join; compute DAS, CEBS, DCI.
Week 2	10 Feb – 14 Feb 2026	Sprint I / Epic 2	Generate 8 EDA figures; identify correlations and outliers.
Week 3	15 Feb – 21 Feb 2026	Sprint I / Epic 3 (part 1)	OLS regression, diagnostics, multiple regression with controls, VIF check.
Week 4	22 Feb – 28 Feb 2026	Sprint I / Epic 3 (part 2)	Robustness suite: bootstrap (10k), HC3, permutation (10k), weight sensitivity, mediation, piecewise regression, extended sample (n = 21).
Week 5	01 Mar – 07 Mar 2026	Sprint II / Epic 4	7 ML models under LOO-CV; XGBoost with nested LOO-CV; 4-fold stability across 400 splits.
Week 6	08 Mar – 15 Mar 2026	Sprint II / Epic 5 + Writing	SHAP beeswarm, bar, and waterfall plots; K-Means (k=2) + PCA biplot; IEEE paper and project report.

CHAPTER 3 — SPRINT PLANNING AND EXECUTION METHODOLOGY

The project follows an Agile / Scrum-inspired execution model with two three-week sprints. Sprint I focuses on data engineering, exploratory analysis, and classical statistical validation. Sprint II focuses on machine learning validation, explainability, and state-typology clustering. Each sprint is documented below with its objectives, functional document, architecture view, outcomes, and retrospective.

3.1 Sprint I — Data Engineering, EDA & Statistical Validation

Sprint Duration: 01 February 2026 – 21 February 2026 (3 weeks)

Sprint Goal: Build a verified 16-state dataset, compute composite indices, generate publication-grade EDA, and produce a defense-ready statistical test of H1, H2, and H3.

3.1.1 Objectives with User Stories of Sprint I

Story ID	User Story	Acceptance / Outcome
S1-US1	As a researcher, I want to ingest all 7 government sources with explicit provenance so the pipeline is reproducible.	01_data_engineering.py produces a merged CSV with no missing values.
S1-US2	As a researcher, I want DAS, CEBS, and DCI computed with documented weight schemes (40/30/30 and 35/40/25).	Three composite-index columns added to the final dataset.
S1-US3	As a researcher, I want 8 publication-ready EDA figures.	Figures fig1–fig8 generated in /figures/.
S1-US4	As a researcher, I want OLS on CEBS ~ DAS with full diagnostics.	$\beta_1 = 0.551$, $R^2 = 0.610$, $p < 0.001$, with Breusch-Pagan, Durbin-Watson, Cook's D.
S1-US5	As a reviewer-proof scientist, I want robustness (HC3, permutation, bootstrap,	Robustness report committed to /results/.

	weight sensitivity).	
S1-US6	As a researcher, I want mediation and threshold tests for H2 and H3.	Sobel z = −0.241; piecewise breakpoint at 118.77% tele-density.

3.1.2 Functional Document

Scope.

Sprint I implements the full data engineering and statistical validation pipeline. It reads raw CSV/Excel files, merges them, derives composite scores, produces EDA artifacts, and executes the statistical battery that supports H1, H2, and H3.

Inputs / Data Sources.

Source (Year)	Variable	Document / Table
TRAI (2024)	Tele-density (%)	Quarterly Indicator Report
RBI (2024–25)	Per Capita NSDP (INR)	Handbook of Statistics, Table 9
NITI Aayog (2023–24)	SDG India Index Score	SDG India Index & Dashboard
NPCI / Dataful (2024–25)	UPI per-capita transactions	Monthly UPI product stats
CPCB (2021–22)	E-waste processed (tonnes)	Annual e-waste report
Census of India (2011)	Literacy rate (%)	Primary Census Abstract
Census of India (2011)	Urban population (%)	Primary Census Abstract

Functional Requirements.

- FR1: Inner-join merge on the State key; fail the build if any source is missing.
- FR2: Min–max normalisation of each raw variable to [0, 100].
- FR3: Compute DAS = $0.40 \cdot \hat{T}d + 0.30 \cdot \hat{U}pc + 0.30 \cdot \hat{N}pc$.
- FR4: Compute CEBS = $0.35 \cdot \hat{E}pc + 0.40 \cdot Ssdg + 0.25 \cdot \hat{N}pc$.
- FR5: Compute DCI = (DAS × CEBS) / 100.
- FR6: OLS of CEBS on DAS with Breusch-Pagan, Durbin-Watson, Cook's distance.

- FR7: Robustness battery — HC3, bootstrap CIs (10k), permutation (10k), weight sensitivity across 7 schemes.
- FR8: Mediation test (Baron & Kenny + Sobel) and piecewise threshold regression.

Non-Functional Requirements.

- NFR1: Reproducibility — fixed random seeds and saved intermediate CSVs.
- NFR2: Traceability — every figure/table maps to a specific script and result file.
- NFR3: Publication-grade outputs — 300 DPI figures.

Key Formulas.

$\hat{x}_i = (x_i - x_{min}) / (x_{max} - x_{min}) \times 100$

$DAS = 0.40 \cdot \hat{T}d + 0.30 \cdot \hat{U}pc + 0.30 \cdot \hat{N}pc$

$CEBS = 0.35 \cdot \hat{E}pc + 0.40 \cdot Ssdg + 0.25 \cdot \hat{N}pc$

$DCI = (DAS \times CEBS) / 100$

$CEBS_i = \beta_0 + \beta_1 \cdot DAS_i + \varepsilon_i$ (primary OLS)

3.1.3 Architecture Document

Sprint I is implemented as a linear, file-first pipeline so every step produces a durable, inspectable artifact. The architecture has four layers: (1) ingestion, (2) merge & feature construction, (3) analysis, and (4) reporting.

Architectural Layer	Responsibility	Implementation / Output
Layer 1 — Ingestion	Reads raw CSV/XLSX for TRAI, RBI, NITI, NPCI, CPCB, Census, Dataful.	01_data_engineering.py
Layer 2 — Merge & Features	Inner-join on State; min–max normalisation; DAS / CEBS / DCI construction.	01_data_engineering.py → /data/trai_state_data_final_v2.csv
Layer 3 — EDA	Correlation heatmap, scatter + OLS, DCI ranking, distributions, UPI trends.	02_eda_visualizations.py → /figures/fig1–fig8

Layer 4 — Statistics	OLS, multiple regression, diagnostics, robustness, mediation, threshold.	03_statistical_analysis.py, 04_reviewer_defense.py → /results/

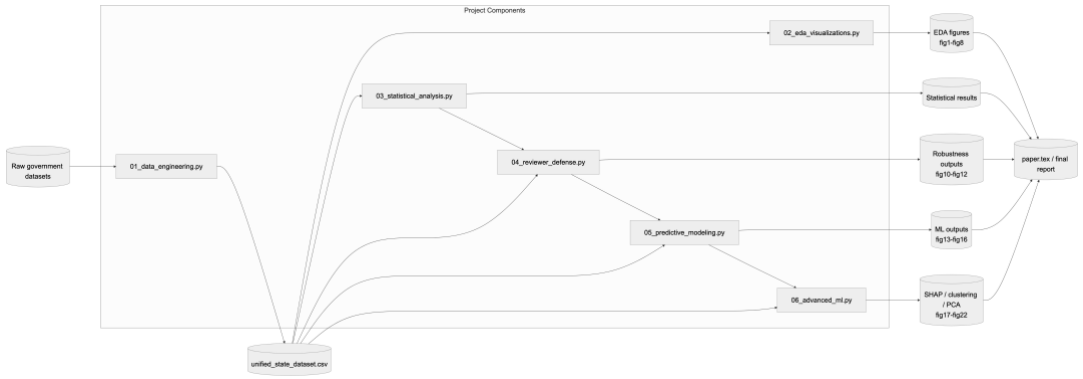


Figure 3.1 High-level system architecture of the research pipeline — ingestion, feature construction, statistical analysis, machine learning, and reporting layers.

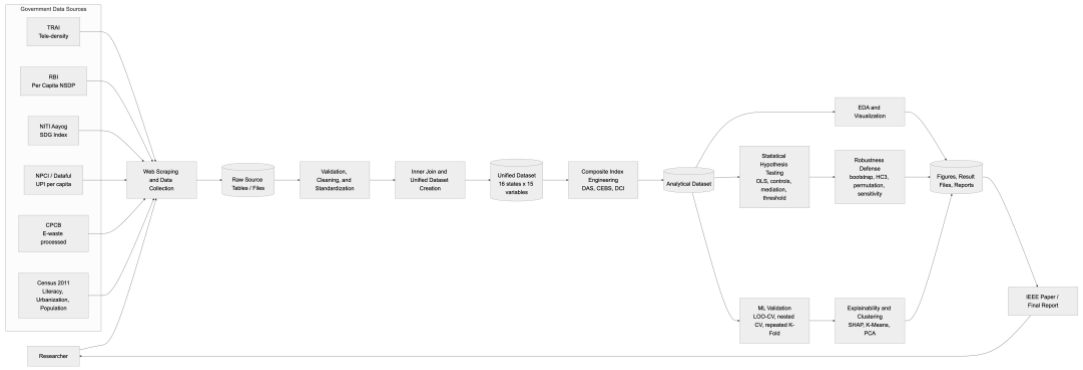


Figure 3.2 Data flow diagram across the seven government data sources feeding the unified 16-state dataset.

Component interactions:

- Each script reads from /data/ and writes to /results/ or /figures/ only — no in-memory state crosses script boundaries.
- Every figure has a matching .txt or .csv results file so numbers in the paper can be traced back to the exact run.
- A fixed random seed (42) is used wherever stochastic resampling is involved (bootstrap, permutation).

3.1.4 Outcome of Objectives / Result Analysis

Descriptive Statistics.

Table 3.3 summarises DAS, CEBS, and DCI for the 16 states retained in the final dataset.

Statistic	DAS	CEBS	DCI
Mean	64.95	46.98	32.52
Std. Dev.	19.69	13.90	17.40
Min	30.27	27.23	8.24
Max	94.40	87.30	72.97
Median	67.02	47.78	32.42

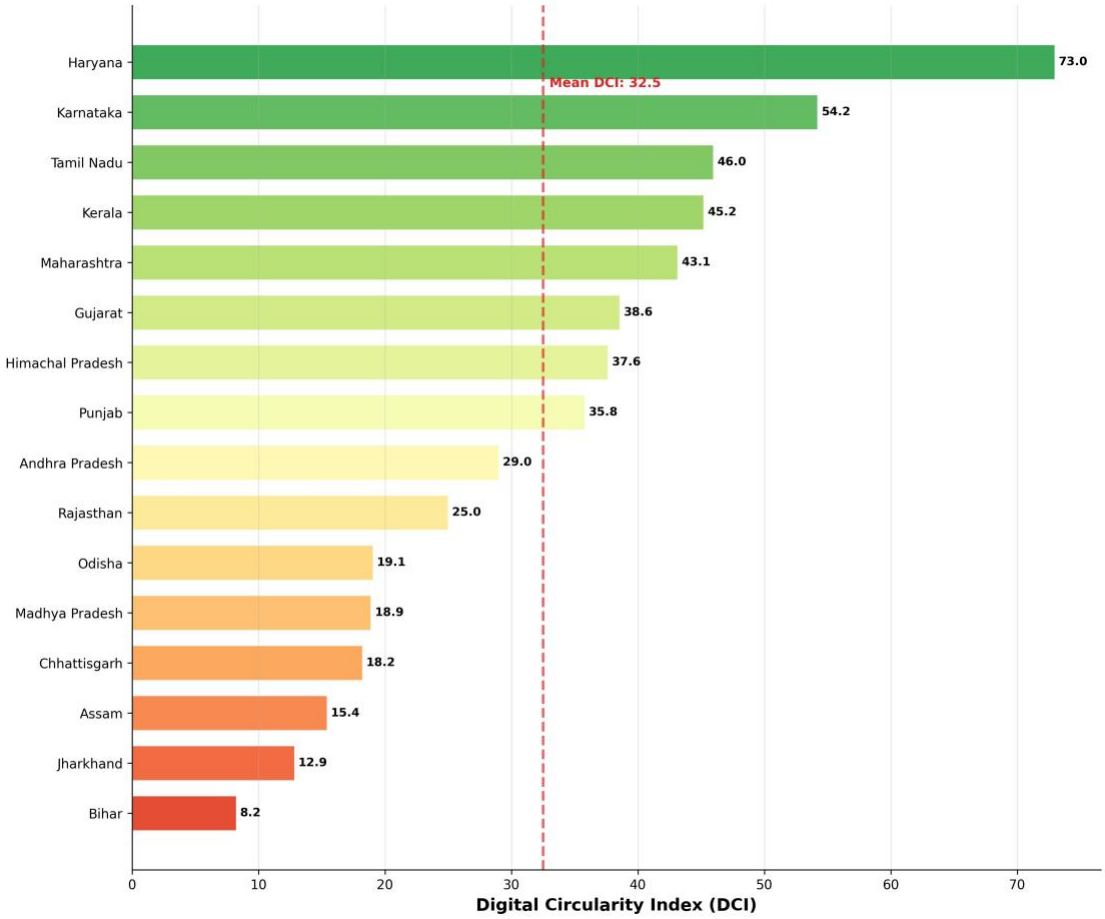


Figure 3.3 DCI ranking across 16 Indian states. Haryana leads (72.97), Bihar lags (8.24).

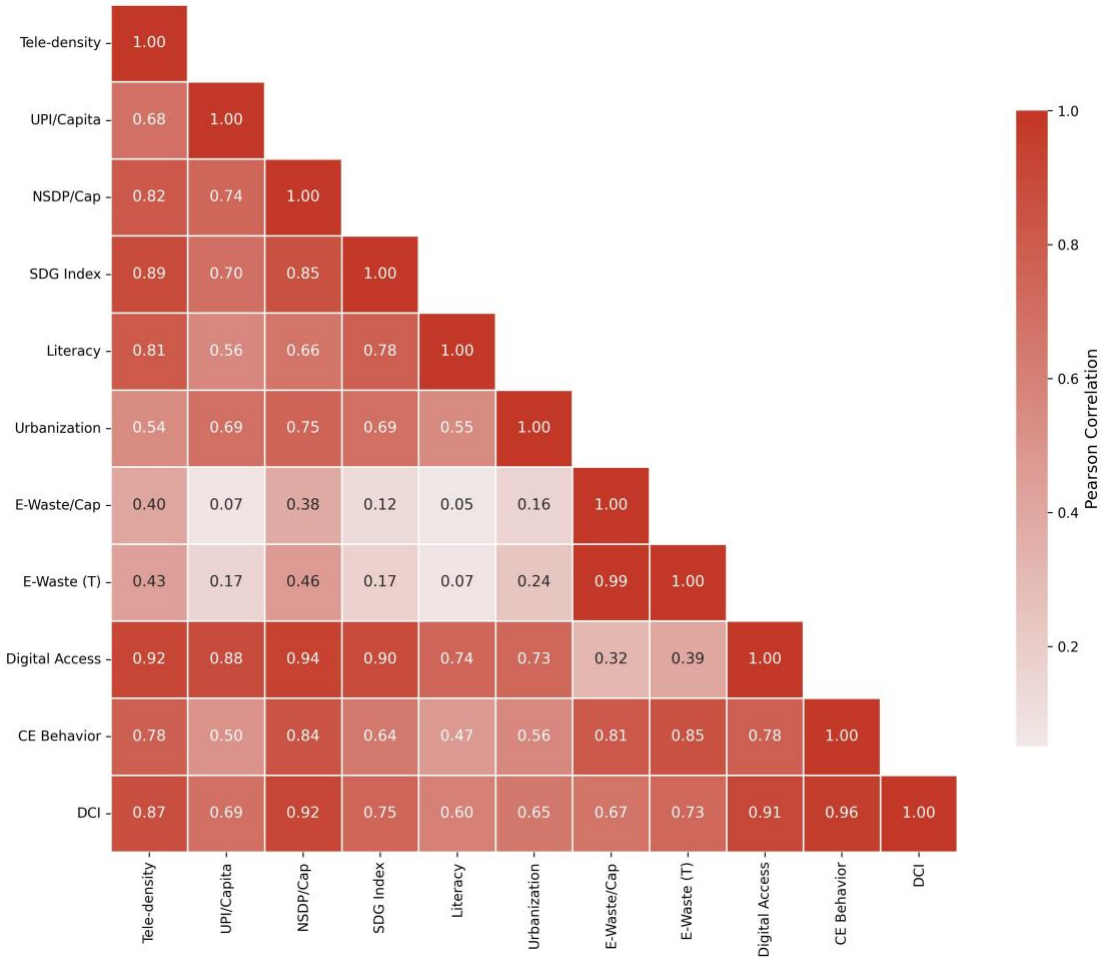


Figure 2.1 Correlation heatmap of all numeric variables. Darker colors indicate stronger correlations.

Primary OLS Result (H1).

The fitted model is: CEBS = 11.158 + 0.551 · DAS. A one-point increase in DAS is associated with ~0.55 points on CEBS, and the model explains 61% of the variance.

Metric	Value
R^2	0.610
Adjusted R^2	0.583
β_1 (slope)	0.551
SE(β_1)	0.118

t-statistic	4.684
p-value	0.000352
F-statistic	21.94
95% CI for β_1	[0.299, 0.804]
Pearson r	0.781

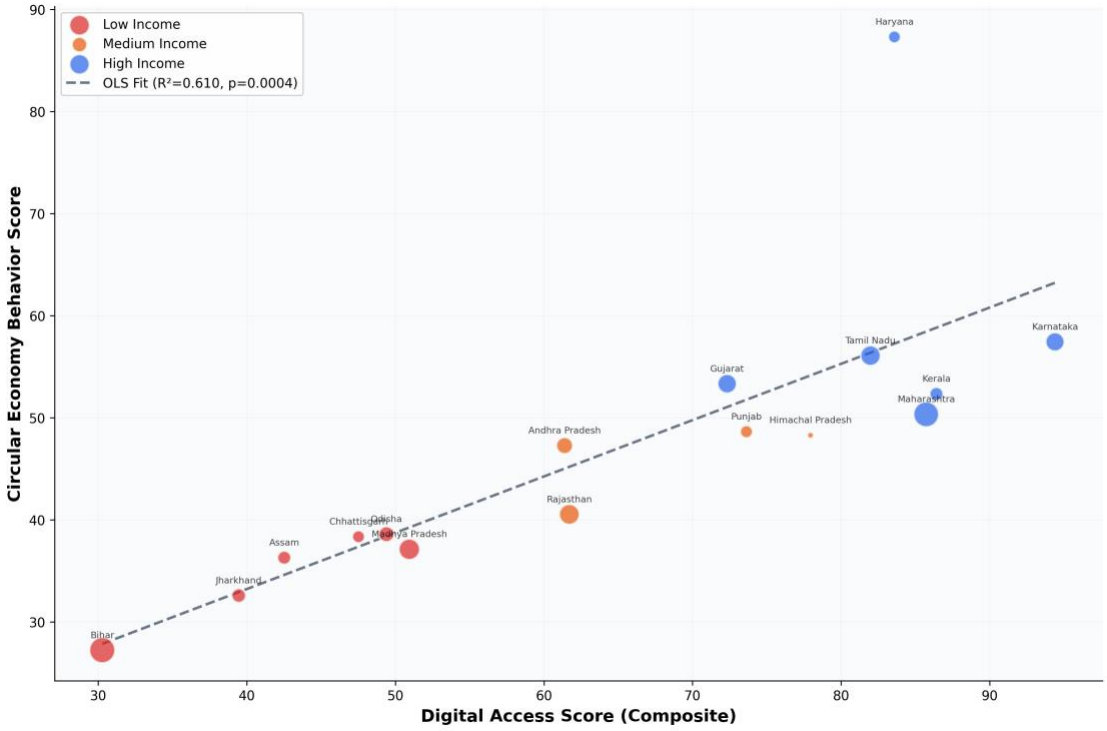


Figure 3.4 Digital Access Score vs. CE Behavior Score (r = 0.781, p < 0.001).

Robustness Diagnostics.

Test	Value	Result
Spearman ρ	0.912	p < 0.001
Kendall τ	0.783	p < 0.001
Shapiro-Wilk	Skew = 2.72	Non-normal
Breusch-Pagan LM	1.539	p = 0.215

Durbin-Watson	2.025	No autocorrelation
Cook's distance	$D_{max} = 4.16$ (Haryana)	1 influential point
Permutation p	0.0001	Significant
HC3 robust p	1.1×10^{-5}	Significant
Bootstrap CI (r)	[0.648, 0.979]	Excludes 0
Bootstrap CI (β_1)	[0.395, 0.836]	Excludes 0

Discussion.

The evidence for H1 is strong and method-robust. A five-control specification lifts R^2 to 0.763 but triggers multicollinearity (VIF 10.57 for DAS, 9.47 for NSDP), so the bivariate model remains the cleaner headline test.

3.1.5 Sprint Retrospective

Category	Notes
What went well	Inner-join strategy produced 100% clean data with zero imputation. EDA figures directly exposed the DAS-CEBS pattern and justified H1. Robustness battery (HC3, permutation, bootstrap) ran end-to-end in one sprint.
What didn't go well	Residual non-normality (Shapiro skew = 2.72) and Haryana's Cook's D = 4.16 required extra defensive analysis. Piecewise regression was unstable above the breakpoint because only 2–3 states sit there.
What we learned	Reporting non-parametric rank tests alongside OLS pre-empts reviewer objections. Documenting weight choices is as important as documenting the model itself.
Action items for Sprint II	(1) Use nested LOO-CV for all ML so hyperparameter tuning is not biased; (2) Report SHAP globally and per-state; (3) Prefer models that degrade gracefully at n = 16 (linear families, RF, XGBoost).

3.2 Sprint II — Machine Learning, Explainability & Clustering

Sprint Duration: 22 February 2026 – 15 March 2026 (3 weeks)

Sprint Goal: Confirm the OLS signal using out-of-sample ML, quantify feature importance using SHAP, and uncover state typologies through K-Means clustering.

3.2.1 Objectives with User Stories of Sprint II

Story ID	User Story	Acceptance / Outcome
S2-US1	As a researcher, I want 7 ML baselines under LOO-CV so I can compare predictive validity.	Full-model Lasso $R^2 = 0.568$; digital-only ElasticNet $R^2 = 0.364$.
S2-US2	As a researcher, I want leak-free nested LOO-CV for XGBoost.	XGBoost $R^2 = 0.511$, $p = 0.002$.
S2-US3	As a researcher, I want stability across 400 splits (100×4 KFold).	RF median $R^2 = 0.603$.
S2-US4	As a researcher, I want SHAP global and per-state explanations.	Tele-density ranked #1 (mean $\|\varphi\| = 5.01$); NSDP #2 (4.88).
S2-US5	As a policy analyst, I want data-driven state clusters.	K-Means (k=2, silhouette-selected) separates Digital Leaders (DCI 46.7) from Emerging States (DCI 18.3).

3.2.2 Functional Document

Scope.

Sprint II consumes the verified dataset from Sprint I and produces out-of-sample validation, explainability artifacts, and state typologies. It is built around three modules: a cross-validation harness, a SHAP explainer, and a K-Means + PCA clustering engine.

Functional Requirements.

- FR9: LOO-CV harness comparing 7 regressors on both digital-only and full feature sets.

- FR10: Nested LOO-CV wrapper (outer LOO, inner 4-fold GridSearch) to remove hyperparameter optimism for XGBoost.
- FR11: Repeated K-Fold CV (100×4 = 400 splits) for stability assessment.
- FR12: SHAP beeswarm, bar, and waterfall plots for all 16 states.
- FR13: K-Means clustering with silhouette selection of k, PCA biplot for visualisation, and cluster profile table.

Models Evaluated.

- Linear Regression, Ridge, Lasso, ElasticNet (linear family)
- SVR with RBF kernel (non-linear baseline)
- Random Forest, Gradient Boosting, XGBoost (tree ensembles)

Key Formulas.

$R^2(LOO) = 1 - \Sigma_i (y_i - \hat{y}_{-i})^2 / \Sigma_i (y_i - \bar{y})^2$

$f(x) = \varphi_0 + \Sigma_j \varphi_j(x)$ (SHAP additive explanation)

3.2.3 Architecture Document

Sprint II is a three-module architecture sitting on top of the Sprint I dataset:

Module	Responsibility	Implementation / Output
Module A — CV Harness	Scikit-learn Pipeline + LeaveOneOut / GridSearchCV for leak-free tuning; writes per-model R^2, RMSE, and Pearson r.	05_predictive_modeling.py, 06_advanced_ml.py → /results/
Module B — SHAP Explainer	shap.Explainer on the tuned XGBoost; generates beeswarm, bar, and waterfall plots for all 16 states.	06_advanced_ml.py → /figures/fig17-19
Module C — Clustering Engine	StandardScaler + KMeans with silhouette scan over k ∈ {2…6}; PCA(2) biplot.	06_advanced_ml.py → /figures/fig20-21

3.2.4 Outcome of Objectives / Result Analysis

LOO-CV Results (7 models).

Model	Digital-Only R^2	Digital-Only r	Full R^2	Full r
ElasticNet	0.364	0.603	0.534	0.731
Lasso	0.347	0.606	0.568	0.758
Ridge	0.338	0.597	0.536	0.743
Random Forest	0.326	0.575	0.490	0.703
Linear Reg.	0.301	0.585	0.442	0.713
SVR (RBF)	0.286	0.554	0.425	0.700
Gradient Boost	0.161	0.542	0.445	0.687

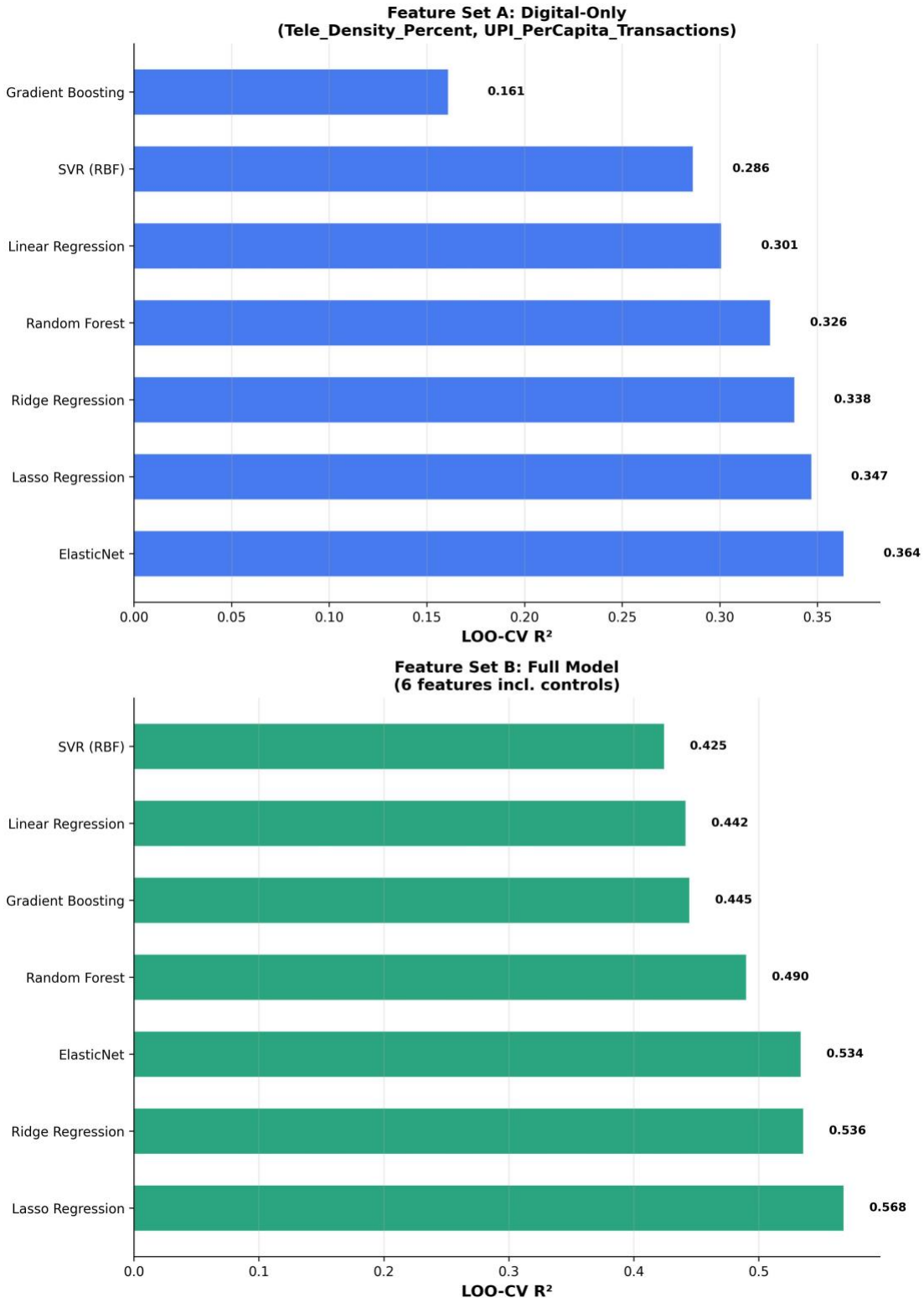


Figure 3.8 ML model comparison — LOO-CV R² across digital-only and full feature sets.

Nested LOO-CV (Leak-Free).

Model	Nested R^2	RMSE	r
XGBoost	0.511	9.41	0.721
Random Forest	0.504	9.48	0.711
ElasticNet	0.471	9.78	0.723
Ridge	0.445	10.03	0.714
Lasso	0.443	10.05	0.713
SVR	−0.023	13.61	−0.593

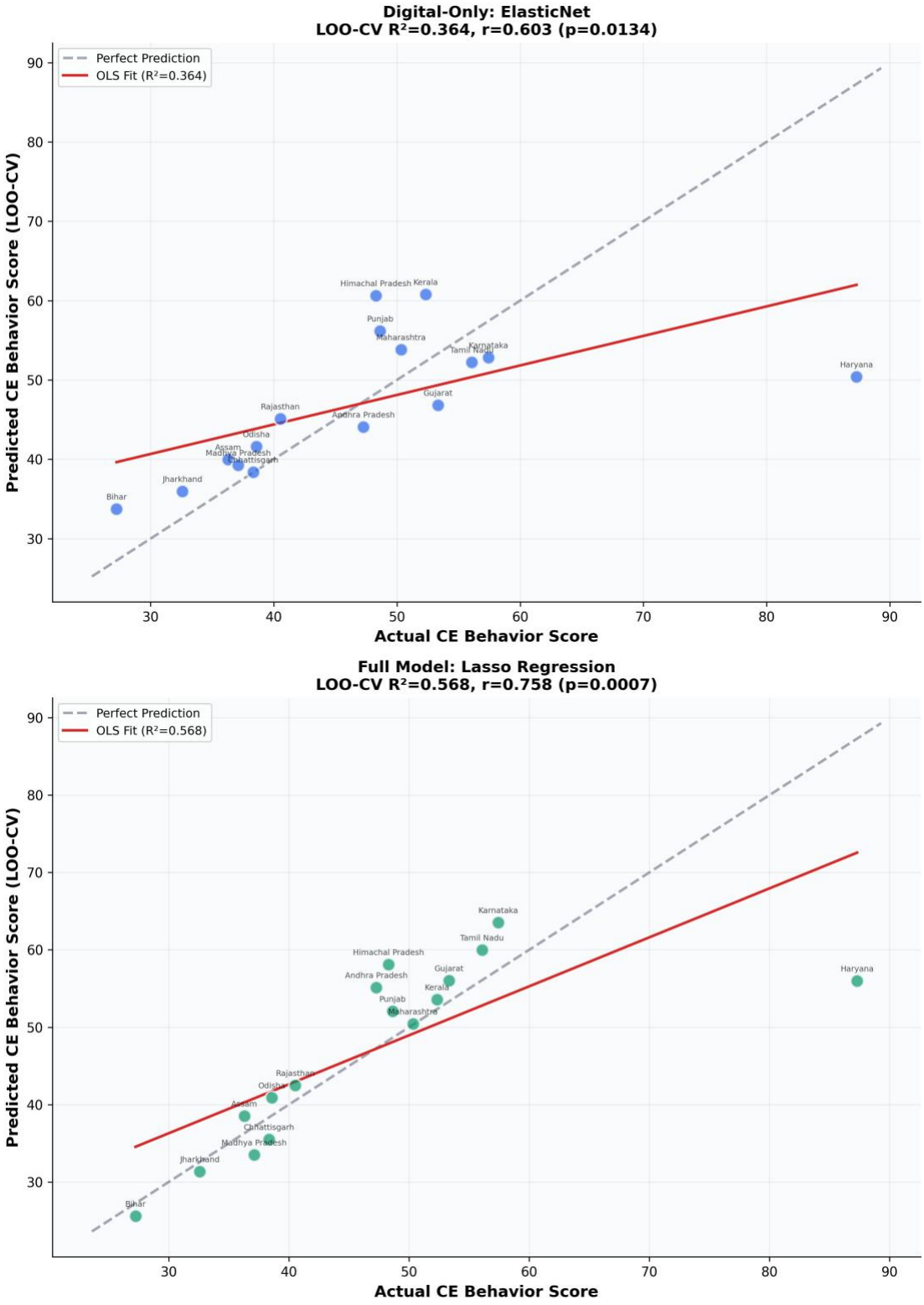


Figure 3.9 Actual vs. predicted CEBS — best nested-CV model.

SHAP Global Feature Importance.

Rank	Feature	Mean |φ|
1	Tele-density (%)	5.012
2	Per Capita NSDP	4.883
3	Population (Mn)	1.714
4	Urban (%)	1.677
5	Literacy Rate (%)	0.998
6	UPI per capita	0.914

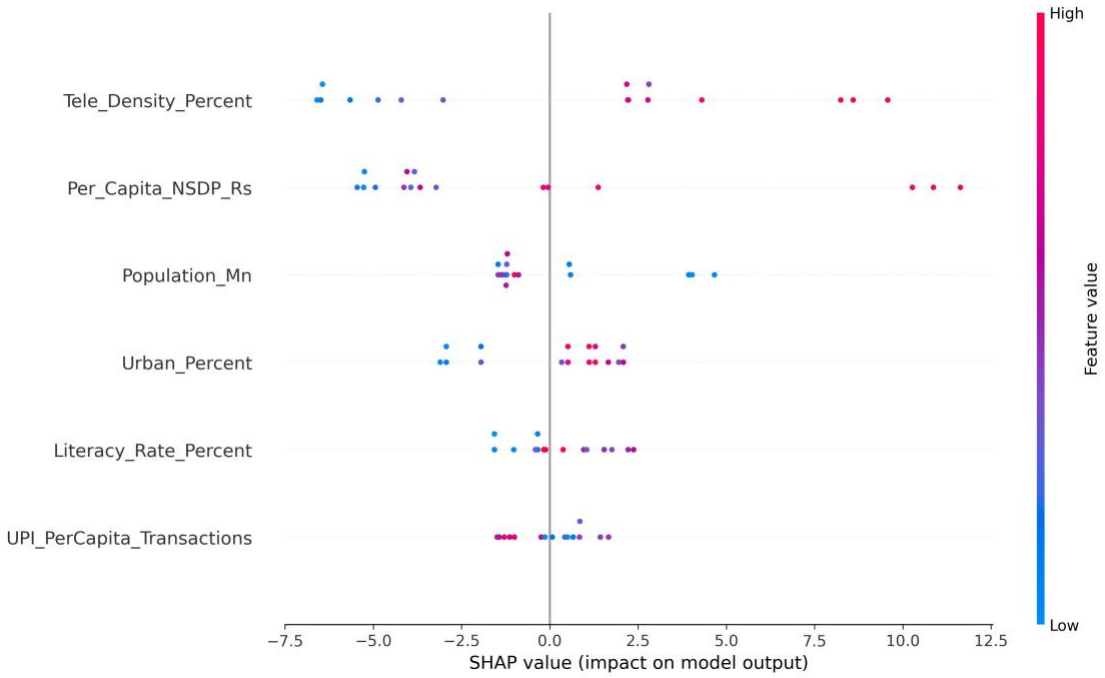


Figure 3.10 SHAP beeswarm plot — each dot is one state-feature pair; red = high value, blue = low value.

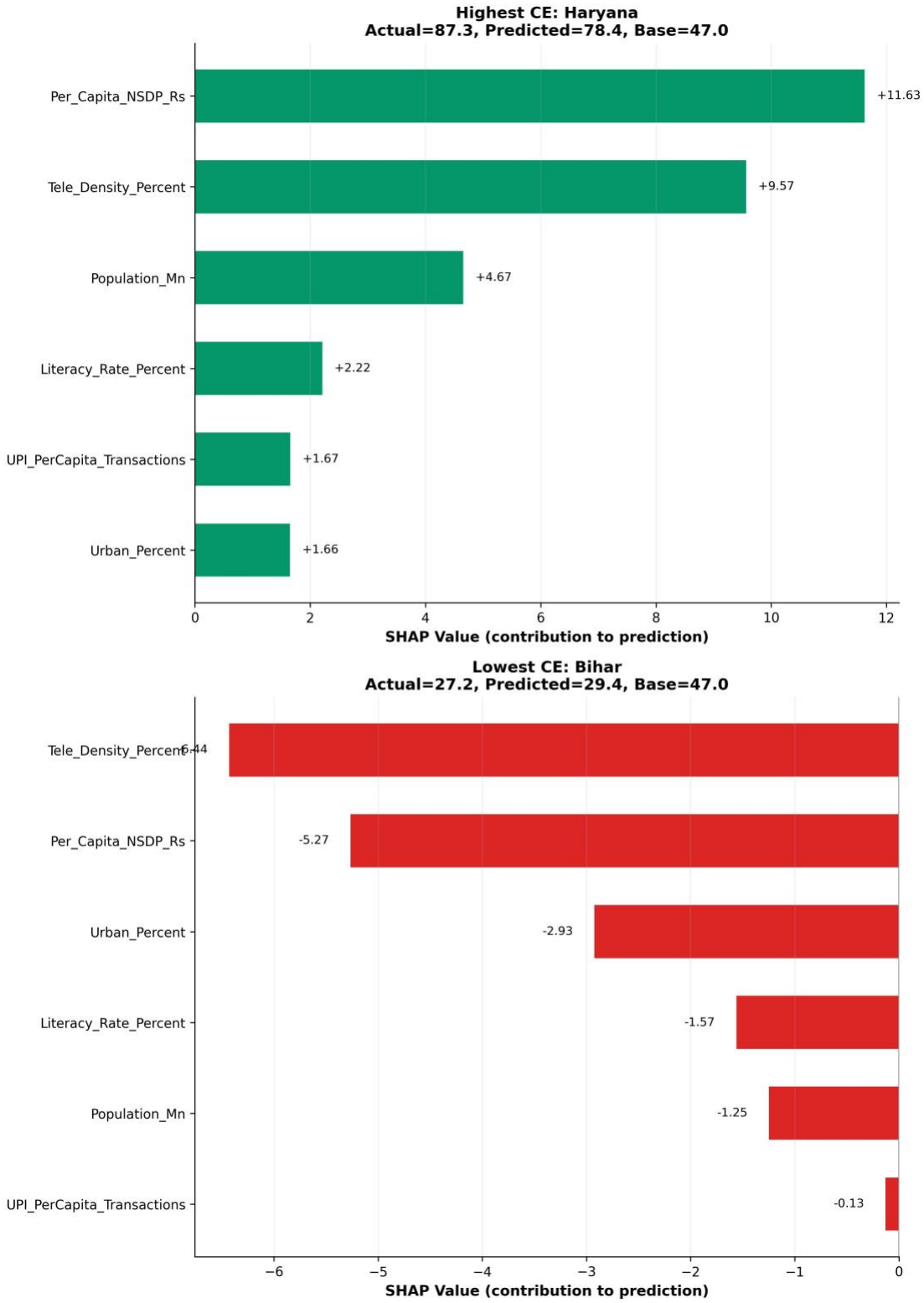


Figure 3.11 SHAP waterfall explanations for Haryana (highest CE) and Bihar (lowest CE).

K-Means Clustering (State Typologies).

Cluster	n	DAS	CEBS	DCI
Digital Leaders	8	82.0	56.7	46.7
Emerging States	8	47.9	37.2	18.3
Gap	—	34.1	19.5	28.4

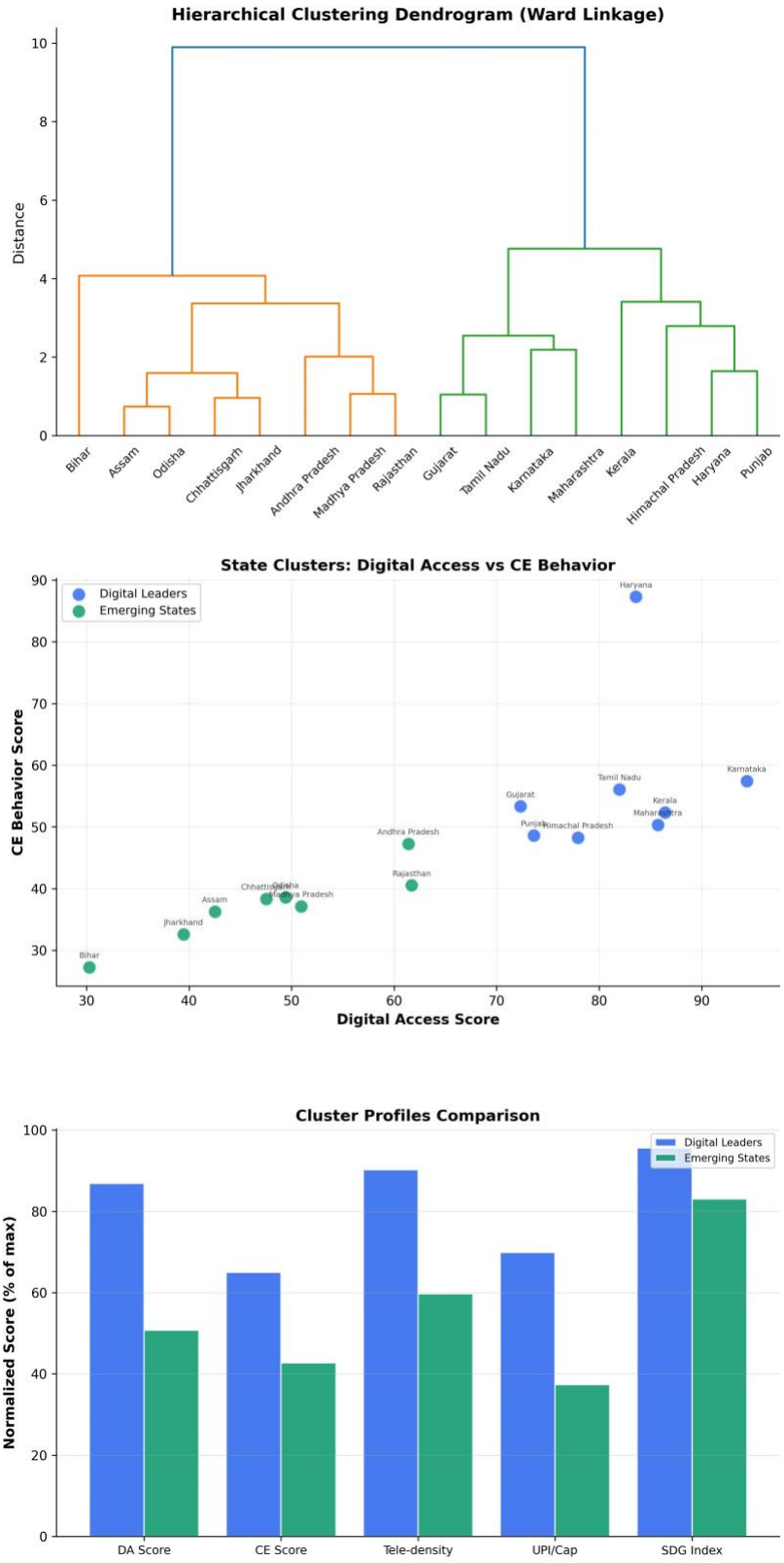


Figure 3.12 K-Means clustering with hierarchical dendrogram and cluster radar profiles. Two distinct state typologies emerge.

3.2.5 Sprint Retrospective

Category	Notes
What went well	Nested LOO-CV delivered a defensible $R^2 = 0.511$ on XGBoost. SHAP results agreed qualitatively with OLS coefficients — tele-density and NSDP as the top drivers. K-Means k = 2 separated states with a 2.5× DCI gap, which translated cleanly into the policy section.
What didn't go well	SVR under nested CV fell below the mean-prediction baseline ($R^2 = -0.023$), highlighting the hazard of non-linear kernels at n = 16. UPI's importance ranked last in SHAP, contradicting an early assumption.
What we learned	At n = 16, regularised linear models and tree ensembles are the only safe choices. Methods should be selected to fail visibly rather than silently. SHAP makes small-sample results interpretable enough for policy audiences.
Action items	(1) Extend validation to 21-state sample by dropping e-waste from CEBS; (2) Report both RF and XGBoost to show that the signal is not a single-model artifact; (3) Add PCA biplot to give reviewers a geometric view of the clustering.

CHAPTER 6 — RESULTS AND DISCUSSIONS

6.1 Project Outcomes (Performance Evaluation, Comparisons, Testing Results)

Integrated performance view.

Combining Sprints I and II, the study produces a multi-method, cross-validated view of the digital-access / CE-behavior relationship. The OLS model gives $r = 0.781$, $R^2 = 0.610$, while leak-free nested-CV XGBoost reaches $R^2 = 0.511$ — the usual in-sample-vs-out-of-sample gap. Table 6 below is the single-glance scorecard used to defend the work.

Evaluation Track	Best Metric	Value	Verdict
Parametric (OLS, bivariate)	R^2	0.610	Supports H1
Parametric (OLS, +5 controls)	R^2	0.763	High but multicollinear
Non-parametric (Spearman ρ)	ρ	0.912	Supports H1
Bootstrap (10,000 resamples)	95% CI for r	[0.648, 0.979]	Excludes 0
Permutation test	p	0.0001	Significant
HC3 robust SE	p	1.1×10^{-5}	Significant
LOO-CV (7 models)	Full-model R^2 (Lasso)	0.568	Confirms out-of-sample
Nested LOO-CV	XGBoost R^2	0.511	Leak-free, p = 0.002
Repeated K-Fold (400 splits)	RF median R^2	0.603	Stable
Extended sample (n = 21)	OLS R^2	0.927	Signal strengthens
K-Means clustering	DCI gap	46.7 vs 18.3 (2.5×)	Two typologies
SHAP global importance	Top feature	Tele-density (5.01)	Drives H1

Mediation Analysis (H2).

Table 6.1 reports the Baron & Kenny / Sobel mediation results. The evidence does not support UPI mediation: the Sobel test is not significant, and the direct effect remains significant after controlling for UPI.

Path	β	p-value	Significant?
c (total): X → Y	0.505	0.0004	Yes
a: X → M	1.499	0.003	Yes
b: M → Y | X	−0.017	0.813	No
c′ (direct): X → Y | M	0.530	0.004	Yes
Indirect (a × b)	−0.026	—	—
Sobel z	−0.241	0.810	No

Threshold Analysis (H3).

The piecewise regression estimates a tele-density breakpoint at 118.77%. Below it, the slope is steep and positive ($\beta_1 = 0.635$, $p = 0.0001$); above it, the slope flips negative ($\beta_2 = -8.738$, $p = 0.047$), but only 2–3 states sit above the breakpoint so we would not over-read the reversal. The piecewise model fits better than the linear one ($R^2 = 0.723$ vs. 0.610), giving qualified support for H3.

Segment	Slope (β)	p-value	Interpretation
Below 118.77% tele-density	0.635	0.0001	Strong positive effect
Above 118.77% tele-density	−8.738	0.047	Reversal (only 2–3 states)
Full piecewise model	$R^2 = 0.723$	—	Better than linear (0.610)

Robustness Visualisations.

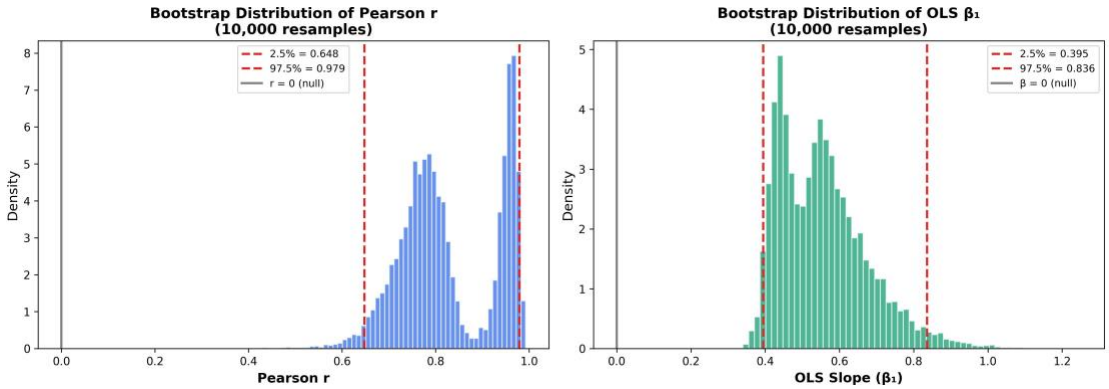


Figure 6.1 Bootstrap distribution of Pearson r (10,000 resamples). The 95% CI [0.648, 0.979] excludes zero.

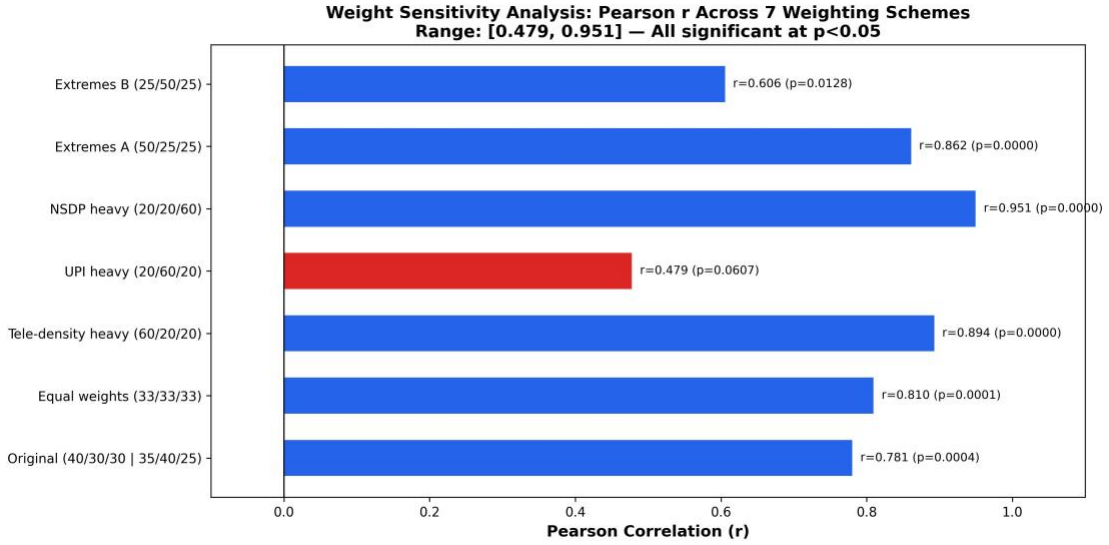


Figure 6.2 Weight sensitivity analysis across 7 alternative DAS weight schemes. Six of seven preserve a significant Pearson correlation.

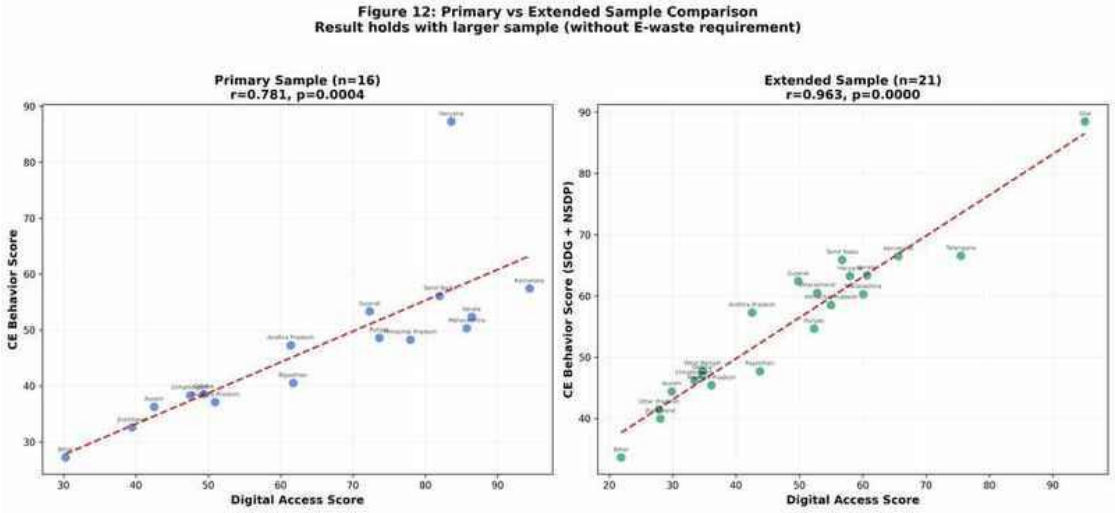


Figure 6.3 Extended sample validation (n = 21, e-waste removed from CEBS). r = 0.963, OLS R^2 = 0.927.

Discussion — do the methods agree?

Broadly, yes. OLS (61% variance explained), nested-CV XGBoost (51%), bootstrap, permutation, and HC3 all point the same direction. For a 16-state sample, that consistency is meaningful evidence against a simple overfitting story.

What drives the relationship?

Random Forest favors NSDP, SHAP favors tele-density, but both agree on the top two drivers. UPI is the unexpected result — smallest SHAP contribution and no mediation effect — suggesting that digital access acts less through payments alone than through broader infrastructure and institutional capacity.

Two Indias in CE performance.

Clustering reveals a sharp divide: one group is strong on both digital access and CE outcomes, the other weak on both. A DCI gap of 46.7 vs. 18.3 is not a mild gradient — it argues for differentiated policy playbooks.

Policy implications.

9. Digital India and Swachh Bharat should be treated less as separate tracks and more as connected systems.

10. Connectivity investment in lagging states should count as environmental capacity-building, not just telecom.
11. Because UPI is not the mediating channel, digital inclusion policy should extend beyond payments to reporting systems, e-governance, and digital literacy.
12. Digital Leaders can move toward advanced monitoring; Emerging States need basic infrastructure first — no single national playbook.
13. SHAP highlights state-specific constraints, suggesting that uniform interventions will miss important differences.

CHAPTER 7 — CONCLUSION AND FUTURE ENHANCEMENT

7.1 Conclusion

This minor project finds a consistent state-level association between digital access and circular-economy behavior in India. Across the 16 states retained in the final dataset, the pattern remains visible under both statistical and machine-learning approaches: the primary OLS model gives $r = 0.781$ ($p < 0.001$, $R^2 = 0.610$), and the leak-free nested-CV XGBoost still reaches $R^2 = 0.511$. Tele-density emerges as a stronger driver than digital payments, and K-Means clustering reinforces the same message by separating states into two groups with a 2.5× DCI gap.

The three hypotheses land as follows: H1 is strongly supported; H2 (UPI mediation) is not supported; H3 receives qualified support with a tele-density breakpoint near 119%. Taken together, the results argue that digital inclusion and circular-economy policy should be treated as connected rather than separate agendas.

7.2 Limitations

- Small sample (n = 16) because only states with complete observations across all seven sources are retained. Haryana's Cook's D = 4.16 shows sensitivity to influential cases, though bootstrap CIs and the 21-state validation reduce concern.
- Cross-sectional design — the study supports association, not causation. Panel data would be needed for stronger causal claims.
- Measurement constraints — literacy and urbanisation still come from Census 2011; NSDP enters both DAS and CEBS; state averages compress within-state variation.
- Composite-index weights are researcher-chosen; while six of seven weight variants still yield a significant result, the choice is not data-driven.

7.3 Future Enhancement

14. Move to panel data and district-level granularity to enable causal designs and capture within-state heterogeneity.

15. Expand the CE index beyond e-waste to include plastic recovery, EPR compliance, material-circularity ratios, and formal recycling capacity.
16. Introduce Bayesian hierarchical models suited for small-n, grouped data.
17. Generalise the framework to other federal economies with comparable administrative data (e.g., Brazil, Indonesia).
18. Build a live policy dashboard that updates DAS, CEBS, and DCI as new government releases arrive, with SHAP explanations per state.
19. Extend explainability to causal SHAP variants and counterfactual policy simulation.

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